# Sunrise Report for Conway, AR During 2024

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April 29, 2025

**Abstract**

This paper analyzes the collection of times of each sunrise during the year 2024 in the town of Conway Arkansas. With this data we can compare the average sunrise of each month during that year. Doing this gives us ab better understanding of the world we live in.

## 1 Introduction

Each year data is collected on sunrise times for a varitery of reasons such as weather forecasting, researching solar events, and for agricuture welfare. This simple project used the data for the sunrise of each day in Conway, Arkansas during the year 2024. The goal of this project is to detail how the data was collected, corrected, and analyzed to find the average sunrise times of each month using python code.

## 2 Analysis

### 2.1 Data Collection

Data was collected from the website timeanddate.com, which tracks the the amount of daylight each day all over the country. Sunrise times were logged in a .csv file in columns for each month. They were logged as strings in the format of hh:mm, for hours and minutes.

Before analysis could begin, data found through (timeanddate.com, 2024) had to be processed before finding the averages of each month could be found. All values of interest represented as strings had to be converted into intergers. This was done by using techniques from (Scopatz & Huff, 2015) to separate each of the strings by the colon use in the values collected for times. From there the minutes had to be coverted to a decimal of the hour, and recombined with the hour the minute are in. With that the data was now able to be analyzed to find the average sunrise time of each month.

### 2.2 Data Aanlysis

With sunrises times from all 365 days in 2024 available the averages of each month were taken 1. This was done using the general formula for calculate mean1.

$$\text{Average} = \frac{\text{Sum of sunrise times in the month}}{\text{Numer of days in the month}} \tag{1}$$

Once this was done for each month the twelve calculated averages were plotted on a graph with sunrise time over months 2.

| | |
|---|---|
| Jan | 7.261828 |
| Feb | 6.920690 |
| Mar | 7.003763 |
| Apr | 6.605556 |
| May | 6.091935 |
| Jun | 5.931667 |
| Jul | 6.134946 |
| Aug | 6.509140 |
| Sep | 6.884444 |
| Oct | 7.282796 |
| Nov | 6.817778 |
| Dec | 7.170430 |

Figure 1: List of the average sunrises for each month in 2024

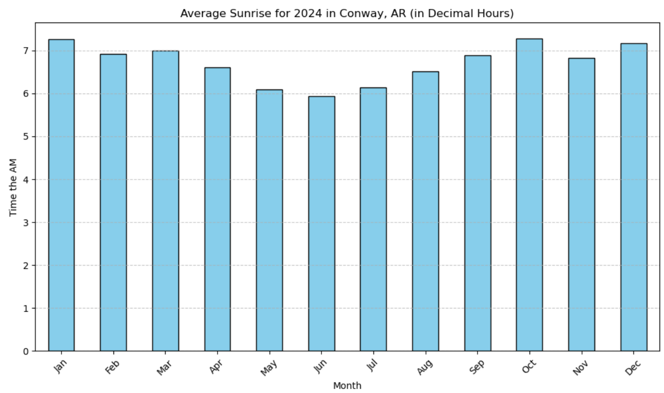


Figure 2: Graph of the sunrise averages for each month in the year 2024

# 3 Conclusion

## 3.1 Results

From figure 2 there is a noticeable dip that appears in the middle of the year. This is due to the rolling back an hour that is done around the United States for daylight savings. The average sunrise time for each month except for June in 2024 was above 6am. For June the average sunrise was at 5:55am 1. For the entire year the average sunrise was at 6:41am. These numbers were noticably effected by the daylight savings time change that is done twice a year by rolling the clock back one hour, then back forward at a later point in the year.

# References

Scopatz, A., & Huff, K. D. (2015). *Effective computation in physics.* O'Reilly Media.

timeanddate.com. (2024). *Conway, arkansas, usa-sunrise,sunset, and daylegnth, 2024.* Retrieved April 29, 2025, from https://www.timeanddate.com/sun/@4106458?month=4&year=2024